Sydney Secondary College
Blackwattle Bay Campus

Practice Examination 2026

YEAR 11
MATHEMATICS EXTENSION 1

Chapters 1 & 2: *Algebraic Techniques; Equations & Inequalities*

General Instructions	Collection
• Writing time — 1 hour 15 minutes • Write using black or blue pen. • Board-approved calculators may be used. **Total — 50 Marks** • All questions may be attempted. **Section I — 10 Marks** • Questions 1–10, 1 mark each. • Circle the correct answer on this paper. **Section II — 40 Marks** • Q11: 15 marks Q12: 15 marks Q13: 10 marks • All necessary working must be shown. • Start each question in a new booklet.	• Write your name, class and teacher on each answer booklet and on your multiple choice answer sheet. • Hand in booklets in a single well-ordered pile. • Hand in a booklet for each question in Section II, even if it has not been attempted. • If you use a second booklet for a question, place it inside the first. • Write your name, class and teacher on this question paper and hand it in with your answers. • Place everything inside the answer booklet for Question Eleven.

NAME:	CLASS:

TEACHER:

Section I — Multiple Choice Answer Grid

Q	A	B	C	D
1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D

SECTION I — Multiple Choice (10 marks)

Circle the letter of the correct answer. Each question is worth 1 mark.

Question 1

Simplify $\dfrac{(3a^2)^3}{9a^4}$.

(A) $3a^2$ (B) $3a^4$ (C) $27a^2$ (D) $9a^2$

Question 2

Which of the following equals $8^{-2/3}$?

(A) -4 (B) 4 (C) $\dfrac{1}{4}$ (D) $-\dfrac{1}{4}$

Question 3

Simplify $\sqrt{50} - 2\sqrt{8} + \sqrt{18}$.

(A) $3\sqrt{2}$ (B) $4\sqrt{2}$ (C) $5\sqrt{2}$ (D) $6\sqrt{2}$

Question 4

Factorise $6x^2 + 7x - 3$.

(A) $(6x-1)(x+3)$ (B) $(3x-1)(2x+3)$ (C) $(2x-1)(3x+3)$ (D) $(6x+1)(x-3)$

Question 5

Solve $|2x+3| = 7$.

(A) $x = 2$ only (B) $x = -5$ only (C) $x = 2$ or $x = -5$ (D) $x = 5$ or $x = -2$

Question 6

The solution to $(x-1)(x+4) \leq 0$ is:

(A) $1 \leq x \leq 4$
(B) $-4 \leq x \leq 1$
(C) $x \leq -4$ or $x \geq 1$
(D) $-1 \leq x \leq 4$

Question 7

The exact solutions of $x^2 - 4x + 1 = 0$ are:

(A) $x = 2 \pm \sqrt{3}$ (B) $x = 4 \pm \sqrt{3}$ (C) $x = 2 \pm \sqrt{5}$ (D) $x = -2 \pm \sqrt{3}$

Question 8

The solution to $2x - y = 5$ and $x + 3y = -1$ simultaneously is:

(A) $x = 2,\ y = -1$ (B) $x = -2,\ y = 1$ (C) $x = 2,\ y = 1$ (D) $x = -1,\ y = 2$

Question 9 *[Extension 1 content]*

Solve $|3 - 2x| < 5$.

(A) $-1 < x < 4$ (B) $x < -1$ or $x >$ 4 (C) $-4 < x < 1$ (D) $-1 \leq x \leq 4$

Question 10 *[Extension 1 content]*

The solution to $\dfrac{4}{x} > 1$ is:

(A) $0 < x < 4$ (B) $x > 4$ (C) $x < 0$ or $x > 4$ (D) $x < 4$

End of Section I

SECTION II — Written Response (40 marks)

Show all necessary working. Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a) Simplify, leaving your answer with positive indices only:

$$\frac{x^{-2} \cdot (2x)^3}{4x^{-1}}$$

2

(b) Evaluate without a calculator:

$$27^{2/3} - 16^{3/4} + 8^{1/3}$$

2

(c) (i) Expand and simplify $(3+\sqrt{5})(3-\sqrt{5})$. **1**

(ii) Hence, rationalise the denominator of $\dfrac{8}{3-\sqrt{5}}$ and simplify your answer fully. **2**

(d) Factorise fully:

(i) $3x^3 - 12x$ **2**

(ii) $2x^2 - 5x - 3$ **1**

(e) Simplify the following expression and state any restrictions on x:

$$\frac{x^2 - 4x + 4}{x^2 - 4} \times \frac{x+2}{x-2}$$

3

(f) Simplify $3\sqrt{48} - 2\sqrt{75} + \sqrt{27}$. **2**

Examination continues next page ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) Solve:

$$\frac{3x+1}{2} - \frac{x-3}{4} = 5$$

2

(b) Solve $-3 < 2x + 1 \leq 7$ and represent the solution on a number line. **2**

(c) Solve $|4x - 3| = 9$. **2**

(d) Solve $2x^2 + 3x - 5 = 0$ by factorisation. **2**

(e) Solve $x^2 - 6x + 7 = 0$ by *completing the square.* Give exact answers in simplest surd form. **2**

(f) Solve $x^2 + 2x - 8 < 0$. **2**

(g) Solve the following simultaneous equations:

$$y = x^2 - 3x \qquad \text{and} \qquad 2x - y = 4$$

3

Examination continues next page . . .

QUESTION THIRTEEN (10 marks) Use a separate writing booklet.

Marks

(a) Solve $2x^2 + 5x - 1 = 0$ using the quadratic formula. Give your answers correct to 3 significant figures. **2**

(b) Solve the following simultaneous equations:

$$y = x^2 - 3x + 4 \qquad \text{and} \qquad y = 2x$$

3

(c) *[Extension 1]* Solve $|2x - 5| \geq 3$. **2**

(d) *[Extension 1]* Solve $\dfrac{5}{x-2} \leq 1$.

Justify your method carefully and state any restrictions on x. **3**

End of Examination